FACULTY OF ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)

B.E. (CSE) IV-Semester (MAIN) Examinations July-2025

ENGINEERING MATHEMATICS – IV

(Probability & Statistics)

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*

ii) Answers must be written in same order as they occur in the Question Paper.

*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	The numbers 3.2, 5.8, 7.9 and 4.5 have frequencies x, (x+2), (x-3) and (x+6) respectively. If arithmetic mean is 4.876, find the value of x.	2	3	1
b)	Write applications of Gamma distribution.	2	2	2
c)	In Poisson distribution 3 $P(x=2) = P(x=4)$ then find $p(x=3)$.	2	3	3
d)	Write normal equations of cubic polynomial.	2	2	4
e)	Define Null and Alternative Hypothesis.	2	1	5
f)	Write the applications of t-distribution.	2	2	5
2. a)	Calculate the measures of central tendency for following data	6	3	1

x	10-15	15-20	20-25	25-30	30-35	35-40	40-45	45-50
f	4	6	11	21	19	8	6	11

b)	A random variable x has the following probability distribution						6	3	1

x	1	2	3	4	5
P(x)	c	c	3c	$c^2 + c$	$6c^2$

Find the value c. evaluate mean and variance of (3X-4)/11, P(X≤2), P(X≥3), and P(I<X≤3).

3. a)	The mean and standard deviation of the wages of 6,000 workers engaged in a factory are Rs. 1,200/- and Rs. 400/- respectively. It distributed normally. Then estimate (a) percentage of workers getting wages above Rs.1,600/- (b) Number of workers getting wages between 600/- and 900/- ($P(0 \le z \le 1) = 0.3413$, $P(0 \le z \le 1.5) = 0.4332$, $P(0 \le z \le 0.75) = 0.02734$)						6	4	2

b) Explain exponential distribution and calculate its variance.

4. a) Calculate the variance and moment generating function of Binomial distribution. 6

b) Define Normal distribution and calculate its moment generating function. 6

5. a) Fit a curve $y = ab^x$ to the following data 6 3

x	-2	-1	1	2	3
y	0.001	0.01	0.538	3.76	26.384

b) In a decathlon held over two days the following performances were recorded in the high jump (x in meters) and long jump (y in meters) Calculate the regression lines 6 3

x	1.90	1.85	1.96	1.88	1.88	0	1.92
y	6.22	6.24	6.50	6.36	6.32	6.44	0

6. a) Fit a Poisson distribution for the following data and test the goodness of fit at 5%level of significance: $(\chi^2_3(0.05) = 7.815)$ 6 3

x	0	1	2	3	4	5
f(x)	110	170	130	60	23	7

b) Measurements of the fat contents of two brands of ice cream A and B yield the following sample data 6 4

Brand A	13.1	13.6	12.8	14.0	14.1	13.9
Brand B	12.7	13.3	14.0	12.9	13.5	13.0

Test the null hypothesis $H_0: \mu_1 = \mu_2$ (where μ_1, μ_2 are the true average fat contents of the ice cream A and B respectively) against the alternate hypothesis $H_1: \mu_1 \neq \mu_2$ test at 5% level of significance. (t=2.228)

7. a) The values in two random samples are given below

Sample1	12	16	18	22	19	28	21
Sample2	10	15	14	19	24	22	

Can we conclude that the two samples are drawn from the same population? Test at 5% level of significance. (F=4.39) 6 3

b) If the density function of a continuous random variable X is given by

$$f(x) = \begin{cases} Kx, & 0 \le x \le 2 \\ (4-x), & 2 \le x \le 5 \\ 0, & Otherwise \end{cases},$$

Find the value of 'K', Mean and $P(0.83 \le x \le 2.75)$ and $P(X > 2.25)$. 6 3

****